

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	35.0	Gender	Male
Height (cm)	162.0	Weight (kg)	85.0
BMI	32.39	Product Info 2	D3
Occupation	prof-specialty	Annual Income	\$50,000.00
Insurance Type	Health	Coverage Amount	\$650,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 3 of 8
Risk Category	Medium Risk
Recommended Annual Premium	\$14,000.00

3. AI Actuarial Analysis

Actuarial Report: Insurance Application Evaluation

Applicant: officer@shield.com

Date: 2023-10-27

1. Risk Assessment

This applicant presents a mixed risk profile.

* **Age (35):** Favorable. Relatively young, typically associated with lower health risks.

* **BMI (32.39):** Unfavorable. This BMI indicates obesity (BMI $\geq$ 30), which is a significant health risk factor, particularly for health insurance. It can lead to various conditions such as cardiovascular disease, diabetes, and musculoskeletal issues.

* **Smoker (0), Alcohol (0), Exercise (1):** Favorable. Non-smoker, no alcohol consumption, and regular exercise are strong positive indicators for health and longevity.

* **Occupation ('prof-specialty'):** Favorable/Moderate. Generally considered a stable, lower-risk occupation compared to manual labor or high-risk professions.

* **Previous Claims (0), Family History (0):** Favorable. Absence of previous claims and a clean family medical history are positive indicators.

* **Income (\$50,000) vs. Coverage (\$650,000):** Potentially unfavorable. The coverage amount represents 13 times the applicant's annual income. While not uncommon for certain life insurance products, for health insurance, this multiple is on the higher side and warrants scrutiny regarding the necessity and potential for over-insurance.

Overall, the positive lifestyle factors (non-smoker, exercise, no alcohol) are strong, but the high BMI is a significant counteracting factor that elevates the health risk.

2. Comparative Analysis

The current application's Predicted Risk Class 3 and Premium of \$14,000.00 fall within the range of the provided historical approved cases, demonstrating a degree of consistency.

* **Application POL-2626 (Similarity: 42.8%, Risk Class: 2, Premium: \$12,000.00):** This is the most similar historical case and has a slightly better risk profile (Risk Class 2) with a lower premium. The current applicant's higher BMI (32.39) is the most probable key differentiator that places them in Risk Class 3 instead of 2, thus leading to a higher premium. Other factors like age, smoking, and previous claims are likely comparable, but the health impact of obesity is significant.

* **Application POL-25755 (Similarity: 40.0%, Risk Class: 5, Premium: \$20,000.00):** This case has a notably worse risk profile than the current applicant, resulting in a higher premium. The current applicant benefits from positive lifestyle factors (non-smoker, exercise, no alcohol) that are likely better than this historical case, despite the high BMI.

* **Application POL-31852 (Similarity: 38.8%, Risk Class: 7, Premium: \$30,000.00):** This case represents a significantly higher risk profile. The current applicant's risk factors are substantially more favorable, explaining the much lower predicted premium and better risk class.

The machine learning model's output places the current applicant appropriately between the low-risk and

high-risk historical examples, reflecting the nuanced balance of positive lifestyle choices and the notable health concern posed by obesity.

3. Anomaly & Fraud Detection

Two potential anomalies are identified:

* **High BMI (32.39):** While not an anomaly in terms of being rare, it is a significant health anomaly for an otherwise healthy profile (non-smoker, exerciser, no alcohol, no family history) and is the primary driver of elevated health risk.

* **High Coverage Amount vs. Income (\$650,000 vs. \$50,000):** A coverage multiple of 13x income is high for health insurance, especially considering the applicant's BMI. This warrants further investigation to understand the specific needs and justification for such a high level of coverage to mitigate potential moral hazard or over-insurance.

No indicators of direct fraud (e.g., inconsistent lifestyle reporting, high claims history coupled with low-risk profile) are immediately apparent. The client identifier "officer@shield.com" is unusual but assumed to be a placeholder for a specific client.

4. Premium Recommendation

The calculated annual premium of **\$14,000.00** appears justifiable:

* **Risk Class 3:** This class indicates a moderate-to-low risk, higher than the best-in-class (Risk Class 2) but significantly better than average or high-risk cases. The premium reflects this positioning.

* **Prediction Probability (55.5%):** The confidence level is moderate. While not exceptionally high, it suggests the model has identified a reasonable risk grouping. The premium likely incorporates a buffer due to this moderate confidence.

* **Historical Comparison:** The premium is \$2,000 higher than the comparable Risk Class 2 case (POL-2626), which is likely attributed to the applicant's high BMI. It is substantially lower than the Risk Class 5 (\$20,000) and Risk Class 7 (\$30,000) cases, aligning with the applicant's overall better profile.

The premium adequately accounts for the positive lifestyle factors while recognizing the increased health risk associated with the applicant's obesity.

5. Underwriting Decision Recommendation

Given the combination of factors, a **Referral for Manual Review** is recommended.

Justification:

1. **High BMI:** While other health indicators are positive, a BMI of 32.39 (obese) for health insurance warrants a closer look by a human underwriter. This could involve requesting recent medical reports, blood tests, or a physician's statement to better understand the current health implications and potential comorbidity risks associated with obesity.

2. **High Coverage Multiple:** The \$650,000 coverage against a \$50,000 income is a significant multiple. An underwriter should assess the applicant's specific needs, financial situation, and justification for this level of health coverage to ensure it is appropriate and not indicative of adverse selection.

3. **Moderate ML Confidence:** A 55.5% prediction probability, while acceptable, leaves room for further scrutiny, especially when coupled with the identified anomalies. A manual review can confirm the risk class and premium.

This approach will allow for a comprehensive assessment of the applicant's health status and financial justification for coverage, leading to a more robust and informed final decision.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$14,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.